



**ApexPlanet Data
Analytics Internship**

Task 4 – Data Storytelling & Statistical Validation

Prepared by: Goli Sravani



ApexPlanet Data Analytics

Project Overview



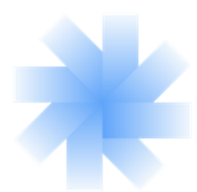
Objective: Analyze sales data to uncover actionable insights and validate findings statistically.

Dataset Overview: The dataset includes key variables such as sales figures, customer records, and order transactions — providing a comprehensive foundation for analysis.

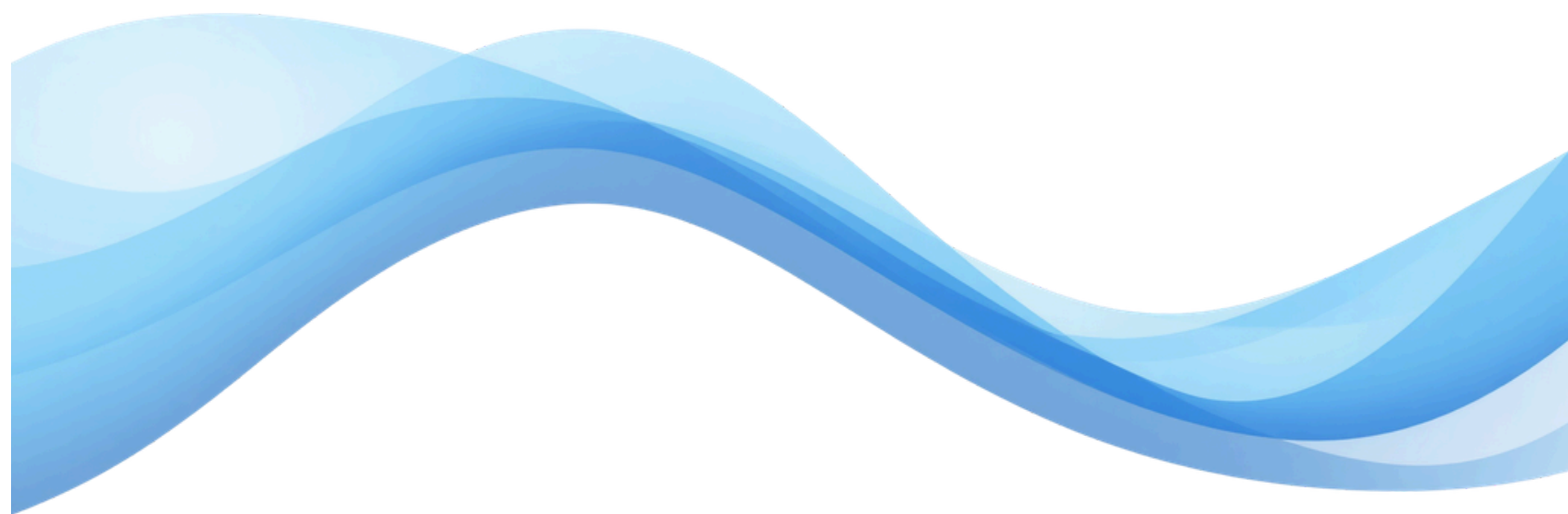
Business Problem: Identify sales trends, customer purchasing behavior, and opportunities to improve business performance through data analysis.

Key Goals:

- Explore and clean the sales dataset
- Identify top-performing categories, cities, and time periods
- Segment customers based on purchasing behavior
- Apply statistical validation to confirm observed trends
- Translate data findings into actionable business recommendations



ApexPlanet Internship



Key Performance Indicators

Total Revenue

₹13.94 Cr

Orders In View

1000

Repeat
Purchase
Rate

5.49%

Average Order
Value

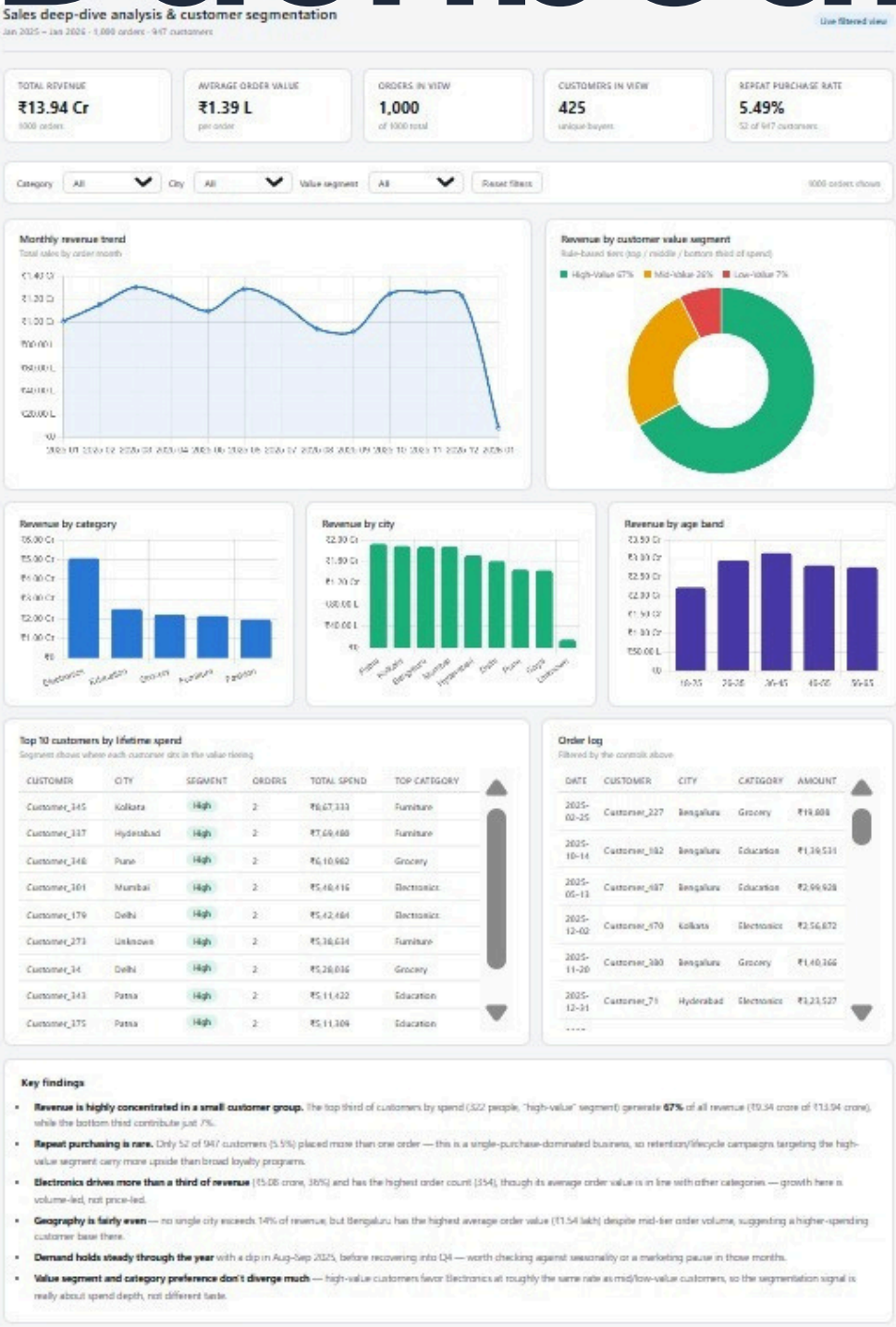
₹1.39 L

Customer
In View

425



Dashboard Overview



RevenueTrend Analysis



Sales by Category



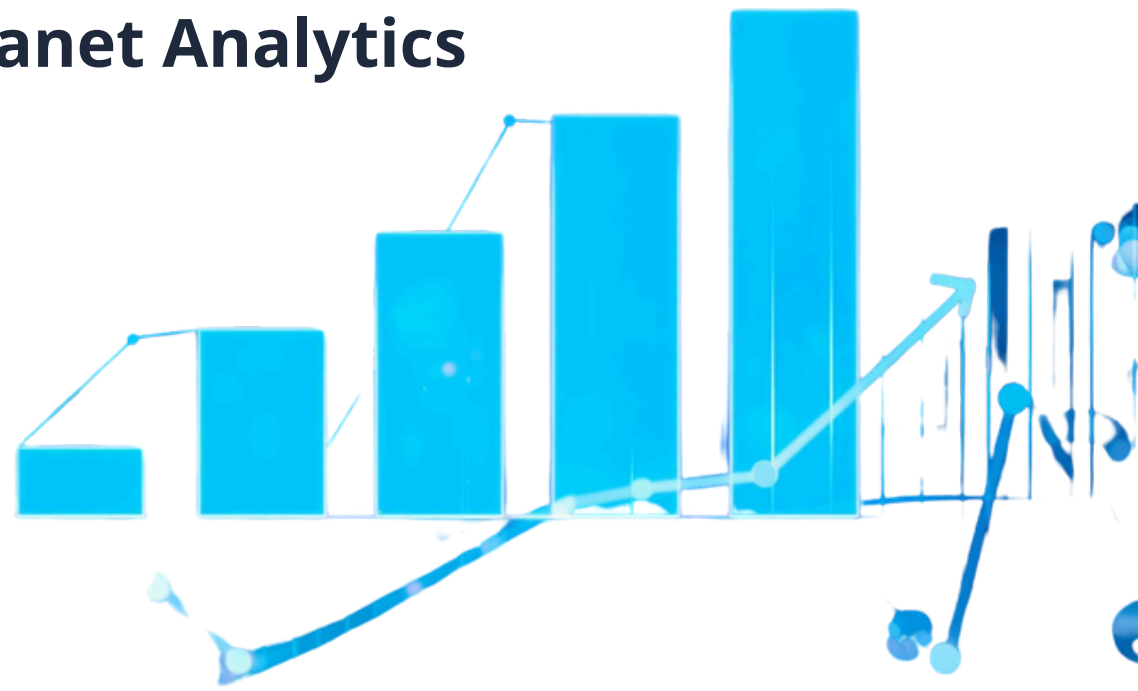
Sales by City



Customer Segmentation



ApexPlanet Analytics



Key Business Insights

Revenue Performance

Total revenue generated:
₹13.94 Cr

Order Performance

Total orders recorded:
1,000

Customer Analysis

Total unique customers:
947

Customer Spending Behavior

Average order value:
₹1.39 L per order

Customer Retention

Repeat purchase rate: 5.49%,



Data Storytelling

01

Business Objective: Improve customer segmentation and optimize sales targeting strategies.

02

Analysis & Findings: Explored sales trends, order patterns, and segment differences across categories and cities.

03

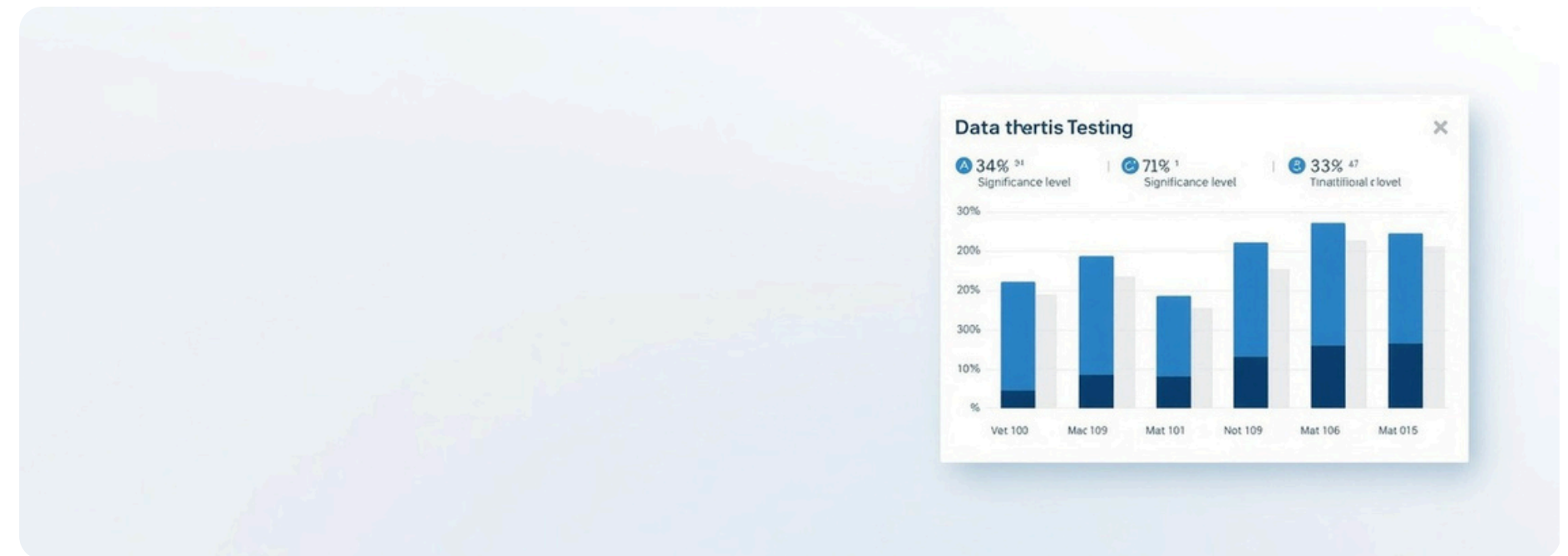
Business Conclusion: Data confirms distinct customer behaviors, enabling targeted strategies for revenue growth.

04

Recommendations: Focus on high-value segments, prioritize top cities, and align promotions with peak sales periods.



Statistical Validation



The statistical results support focusing marketing efforts on high-performing customer segments.

Null Hypothesis (H_0)

No significant difference
in sales between
customer groups

Alt. Hypothesis (H_1)

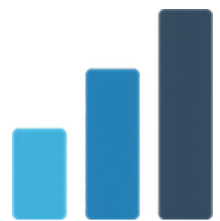
Significant difference
exists between groups

Testing Approach

Independent t-test & One-Way
ANOVA applied across
segments

Results & Conclusion

p-value < 0.05; H_0 rejected.
Significant sales variation
confirmed across segments.



ApexPlanet Data Analytics Internship

Thank You!

Skills Gained: Data Analysis · Visualization ·
Storytelling · Statistical Validation

Prepared by: Goli Sravani